Data Visualization



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With assistance from AI tools, particularly Microsoft Copilot and Google Gemini

🧭 Eight Analytical Purposes

1 Trend

How does it change over time?

Line chart

2 Comparison

How do categories differ?

Bar chart

3 Ranking

What is highest or lowest?

Sorted bar

4 Part-to-Whole

How does each part contribute?

Stacked bar

5 Distribution

How are values spread?

Histogram / box

6 Relationship

How are variables related?

Scatter plot

7 Geographic

Where does it happen?

Choropleth map

8 Target / Deviation

How do we compare with plan?

Actual-vs-target bar

Each will be covered:

question → chart options → example → how to read it → best practices → code

A five-step process

1. Question	What decision needs an answer?	Are we growing, and where?
2. Visualization	Which chart answers it?	Monthly revenue line chart
3. Observation	What does the chart show?	Revenue rose from 120k to 175k.
4. Insight	What does the pattern mean?	Growth accelerated after May, the metric increased faster each month.
5. Implication	Why does it matter, what next?	Plan Q3 capacity for the higher demand level, and verify whether the pricing change drove the acceleration.

🧠 Observation vs insight vs implication

OBSERVATION

What the chart shows

"Online revenue grew 46% between January and June, while Retail was flat."

***Descriptive**: verifiable from the chart alone · no domain knowledge required.*

INSIGHT

What the pattern means

"Growth is coming entirely from one channel, so the business is becoming dependent on Online."

***Interpretive**: compares the pattern to an expectation · needs context.*

IMPLICATION

Why it matters, and what to do

"Concentration risk is rising: protect Online margins and test a second growth channel this quarter."

***Decision-oriented**: names an action or a risk.*

👁 Why visualization is essential to analysis

Three Core Reasons:

- **It’s Faster**: Visuals reveal patterns, such as trends, clusters, and outliers, much faster than tables of numbers.
- **It Reveals the Data**: Summary statistics can hide important differences. A plot shows the actual shape and patterns in the data.
- **It Supports Thinking**: Plotting data helps you explore, question, and discover insights - not just present results.

Motto: Visualize Before Analyzing!

- Charts reveal what summary statistics hide.
- Spot patterns, distributions, outliers, and relationships fast.
- *Anscombe's quartet*: four datasets with identical mean, SD, and regression line - yet completely different shapes.
  - Always plot the data before you trust the statistics.



`06_visualization\00_why_visualization_anscombe.ipynb`

📈 Purpose 1 - Trend

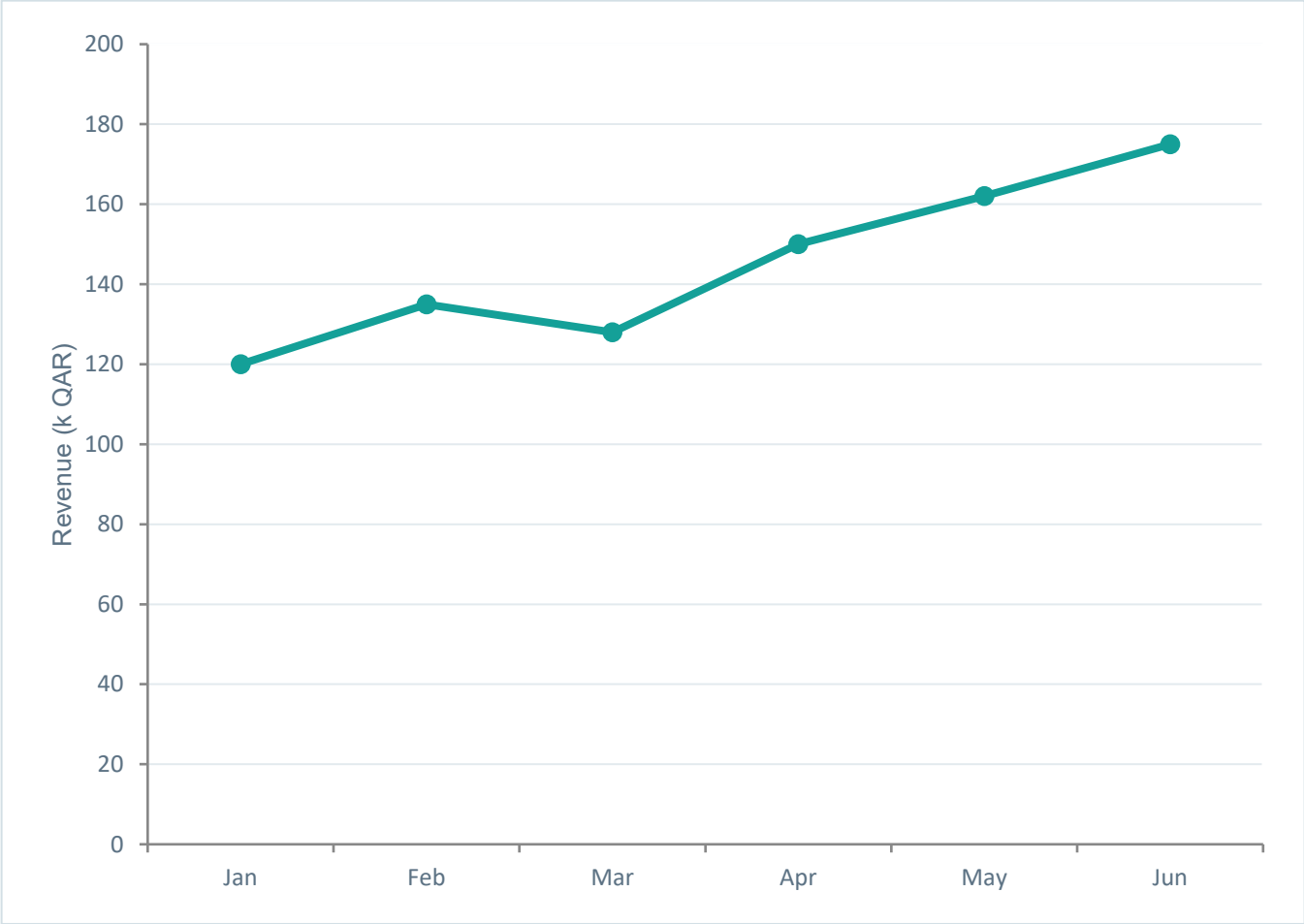
Analytical Question

How does the metric change over time?

Recommended charts - and when to use each

★ DEFAULT - LINE CHART	ALTERNATIVE - AREA CHART	ALTERNATIVE - BAR CHART
Shows slope, turning points and seasonality. Use for 5+ time points.	Use when the size of a total matters. Avoid overlapping areas.	Use when periods are few and discrete, e.g. 4 quarters.

📈 Trend - example and interpretation



ShopStream monthly revenue, Jan–Jun

How to read it

Steeper **slope** means faster change.

Check the y-axis starts at zero.

What to look for

Direction, rate of change, turning points, seasonality, and any flat or missing segment.

✓ Insight

“Revenue rose 46% over six months, with the fastest growth between April and June.”

✕ Not justified

“Revenue will reach 250k by December.” Six points cannot support an extrapolation, and no seasonal cycle has been observed yet.

📈 Trend - best practices

- Keep time on the x-axis, evenly spaced, oldest on the left.
- Start the value axis at zero for amounts; say so when you deviate.
- Limit to 3–5 lines; highlight one and mute the rest.
- Label the final point directly instead of using a legend.
- State the aggregation level in the title: daily, weekly, monthly.
- Break the line where data is missing, and exclude or shade partial periods.

💻 Trend - implementation in Seaborn and Plotly

Seaborn · static, report-ready

```
import pandas as pd, seaborn as sns
import matplotlib.pyplot as plt

df = pd.DataFrame({
    'month': ['Jan','Feb','Mar','Apr','May','Jun'],
    'revenue': [120, 135, 128, 150, 162, 175]})

sns.set_theme(style='whitegrid')
ax = sns.lineplot(data=df, x='month', y='revenue',
                  marker='o', linewidth=2.5)
ax.set_ylim(0, 200)
ax.set(xlabel='', ylabel='Revenue (k QAR)')
ax.set_title('Revenue grew 46% in H1 2025')
plt.show()
```

Plotly · interactive, dashboard-ready

```
import pandas as pd
import plotly.express as px

df = pd.DataFrame({
    'month': ['Jan','Feb','Mar','Apr','May','Jun'],
    'revenue': [120, 135, 128, 150, 162, 175]})

fig = px.line(df, x='month', y='revenue',
 markers=True, title='Revenue grew 46% in H1 2025')
fig.update_yaxes(range=[0, 200],
                 title='Revenue (k QAR)')
fig.update_xaxes(title='')
fig.show()
```

Seaborn returns a static Matplotlib chart → ideal for reports and PDFs. Plotly returns an interactive figure with hover values → ideal for dashboards.

Purpose 2 - Comparison

Analytical Question

How do categories differ from one another?

Recommended charts - and when to use each

★ DEFAULT - BAR CHART

Bar length from a **zero baseline** encodes the value — the most accurate visual comparison. Sort bars by value and go horizontal when labels are long.

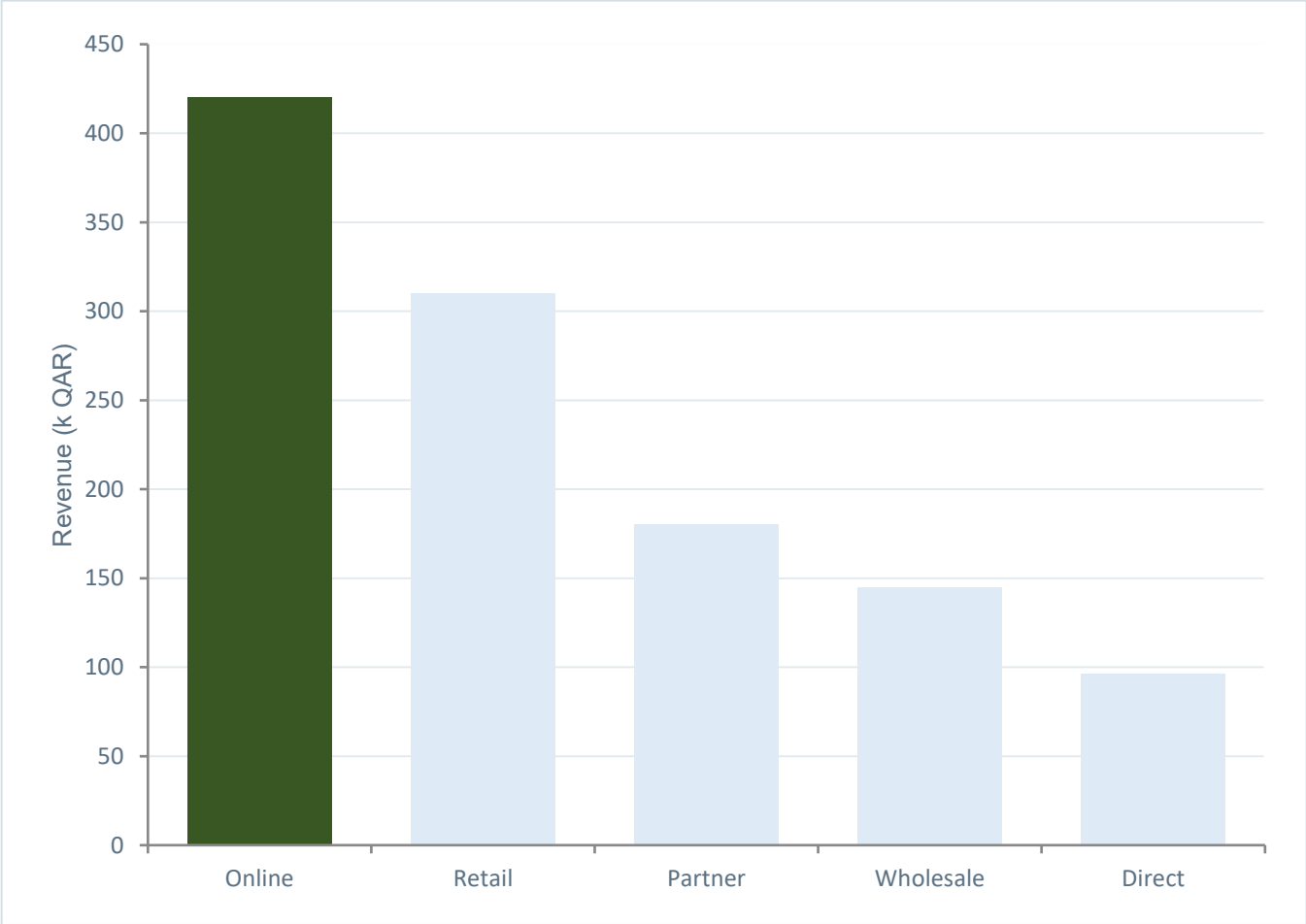
GROUPED BAR CHART

Use when each category splits by a second variable (region × quarter).

Keep to 2–3 groups per category and under ~12 categories.

Practical rule: Start axis at zero · Use one color, a second only for emphasis · Compare percentages when group sizes differ · Avoid 3D bars

Comparison - example and interpretation



Revenue by sales channel, H1

How to read it

Compare bar lengths from a common zero baseline; longer bars represent higher values.

What to look for

Which categories lead, the size of the gap, and whether a few categories dominate the total.

✓ Insight

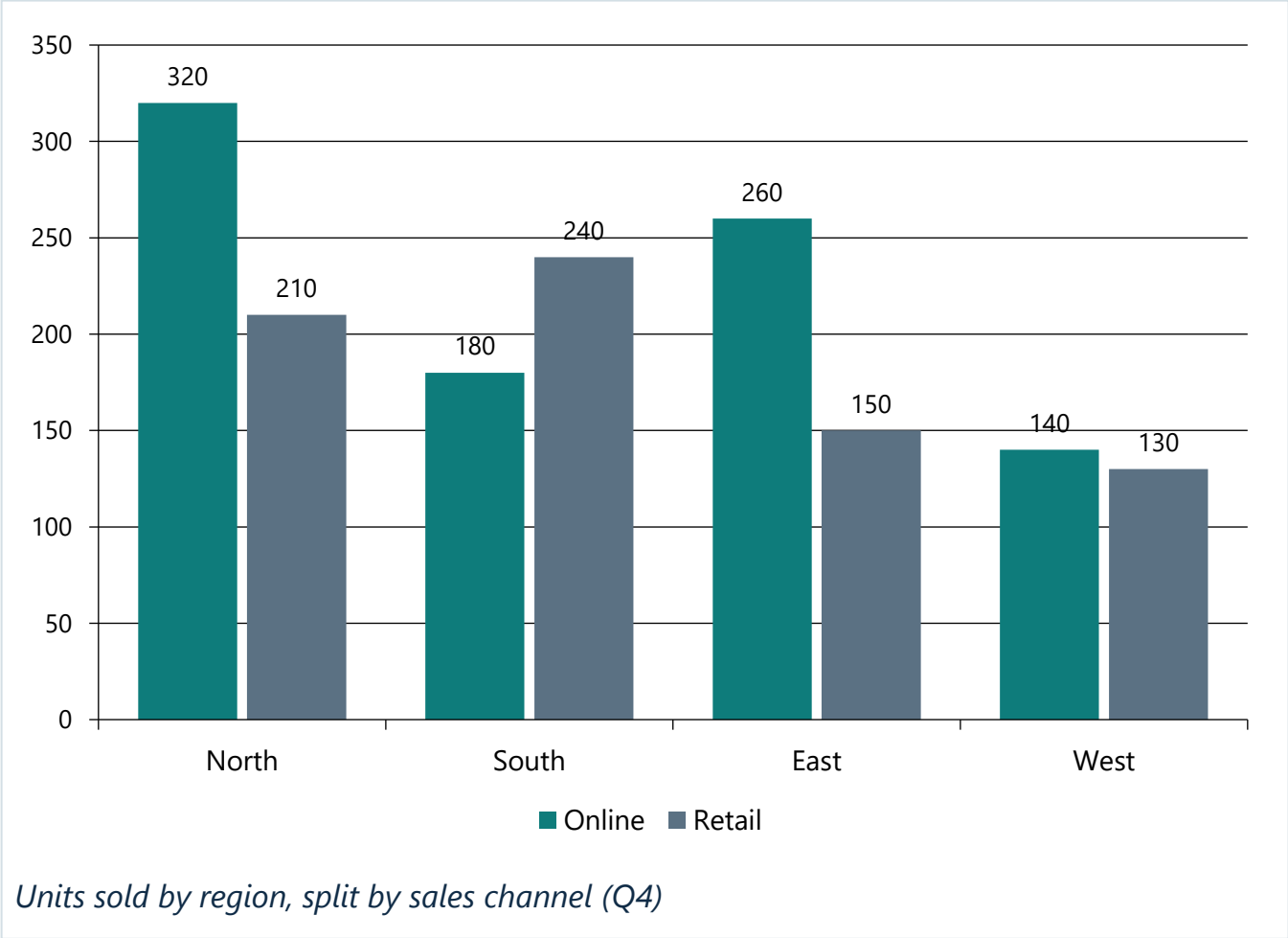
Online generates 4.4× the revenue of Direct and more than a third of H1 revenue.

✕ Not justified

Online is our most profitable channel. Profitability cannot be inferred from this chart without accounting for acquisition costs and other expenses.

📊 Grouped bar chart

Compare subgroups side by side across categories



Units sold by region, split by sales channel (Q4)

When to use it

One **main category**, such as region split by a small number of **subgroups**, such as sales channel. Keep the split to 2–3 groups and the categories under ~12.

How to read it

Compare bars within a region to evaluate the channel mix.
Compare bars of the same color across regions to evaluate regional performance for each channel.

✓ Insight

Online leads in the North, East, and slightly in the West, while Retail leads in the South. Channel performance therefore differs by region.

✕ Avoid

Too many categories or subgroup bars
Adding a third grouping variable
Using stacked bars when precise subgroup comparisons are required

Comparison - best practices

- **Start the value axis at zero** because bar length represents the value.
- **Sort bars by value**, unless categories have a natural order.
- **Use horizontal bars for long labels** and limit the number of categories.
- **Use one color by default** and a second color only for emphasis.
- **Compare rates or percentages**, not raw counts, when group sizes differ.
- **Avoid 3D effects** because they distort perceived bar lengths.

💻 Comparison - implementation in Seaborn and Plotly

Seaborn · static, report-ready

```
import pandas as pd, seaborn as sns
import matplotlib.pyplot as plt

df = pd.DataFrame({
    'channel': ['Online','Retail','Partner',
                'Wholesale','Direct'],
    'revenue': [420, 310, 180, 145, 96]})

sns.set_theme(style='whitegrid')
ax = sns.barplot(data=df, x='channel',
                 y='revenue', color='#14A098')
ax.set(xlabel='', ylabel='Revenue (k QAR)')
ax.set_title('Online delivers 4.4x the revenue of
Direct')

plt.show()
```

Plotly · interactive, dashboard-ready

```
import pandas as pd
import plotly.express as px

df = pd.DataFrame({
    'channel': ['Online','Retail','Partner',
                'Wholesale','Direct'],
    'revenue': [420, 310, 180, 145, 96]})

fig = px.bar(df, x='channel', y='revenue',
             text_auto=True,
             title='Online delivers 4.4x the
                    revenue of Direct')
fig.update_traces(marker_color='#14A098')
fig.update_layout(yaxis_title='Revenue (k QAR)',
                  xaxis_title='')
fig.show()
```

Note the single color in both libraries. Default palettes assign a different color per bar.

🏆 Purpose 3 - Ranking

Analytical Question

What is highest, what is lowest, and in what order?

Recommended charts - and when to use each

★ DEFAULT - SORTED HORIZONTAL BAR

Sorting does the analytical work

Horizontal fits long names.

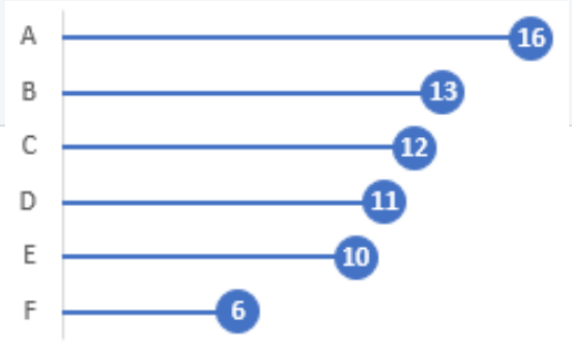
ALTERNATIVE - TOP-N CHART

Use when only leaders matter. Always add an 'Other' bar.

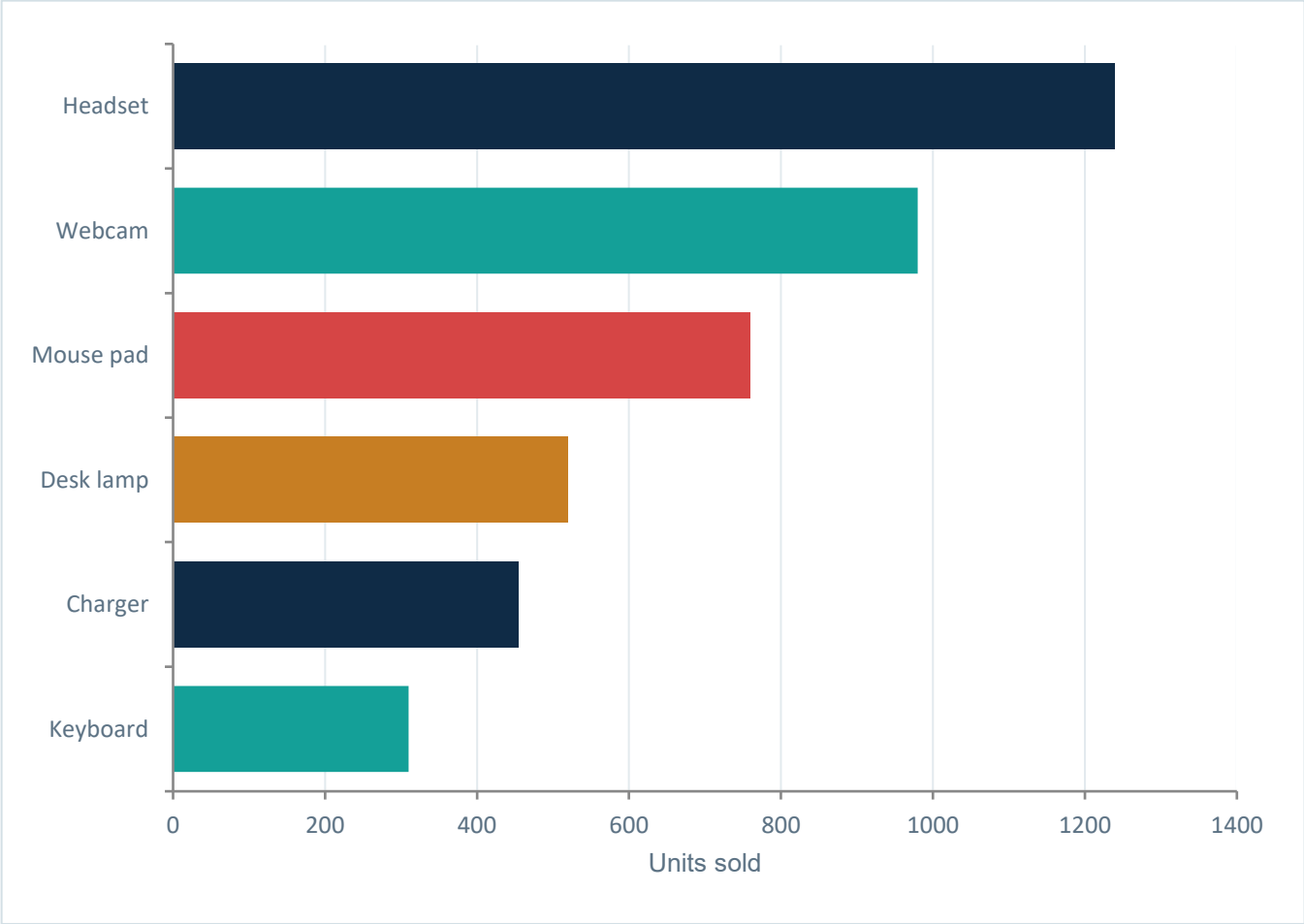


ALTERNATIVE - LOLLIPOP CHART

Same information, less ink, for 15–30 items.



🏆 Ranking - example and interpretation



Top six products by units sold, Q2 (sorted)

How to read it

Read from the longest bar down. The order is the message; gaps show how big the lead is.

What to look for

The leader, the size of the gap to second place, and where the curve flattens into a long tail.

✓ Insight

“Headsets lead on volume at 1,240 units, 27% ahead of webcams; the top three account for most unit sales.”

✕ Not justified

“Headsets are our best product.” Units are not revenue or margin - a cheap high-volume item can be the least valuable.

🏆 Ranking - best practices

- Sort descending; never leave the default alphabetical order.
- Use horizontal bars so names read naturally.
- Show the full ranking, or add an 'Other' bar and state how many were cut.
- Highlight only the item the narrative is about.
- Add value labels when exact figures drive the decision.
- Rank on a stable, decision-relevant metric and treat tiny gaps as noise.

💻 Ranking - implementation in Seaborn and Plotly

Seaborn · static, report-ready

```
import pandas as pd, seaborn as sns
import matplotlib.pyplot as plt

df = pd.DataFrame({
    'product': ['Headset','Webcam','Mouse pad',
                'Desk lamp','Charger','Keyboard'],
    'units': [1240, 980, 760, 520, 455, 310]})
df = df.sort_values('units', ascending=False)

sns.set_theme(style='whitegrid')
ax = sns.barplot(data=df, y='product', x='units',
                 orient="h", color='#C77E23')
ax.set(ylabel='', xlabel='Units sold')
ax.set_title('Headsets lead Q2 volume, 27% ahead of
webcams')
plt.show()
```

Plotly · interactive, dashboard-ready

```
import pandas as pd
import plotly.express as px

df = pd.DataFrame({
    'product': ['Headset','Webcam','Mouse pad',
                'Desk lamp','Charger','Keyboard'],
    'units': [1240, 980, 760, 520, 455, 310]})

fig = px.bar(df, x='units', y='product',
orientation='h', title='Headsets lead Q2 volume')
fig.update_yaxes(categoryorder='total ascending')
fig.update_traces(marker_color='#C77E23')
fig.show()
```

Sort: Pandas **sort_values** in Seaborn, versus Plotly's **categoryorder='total ascending'**.

Use 'ascending' for horizontal bars because Plotly draws the first category at the bottom.

Purpose 4 - Part-to-Whole

Analytical Question

How does each part contribute to the total?

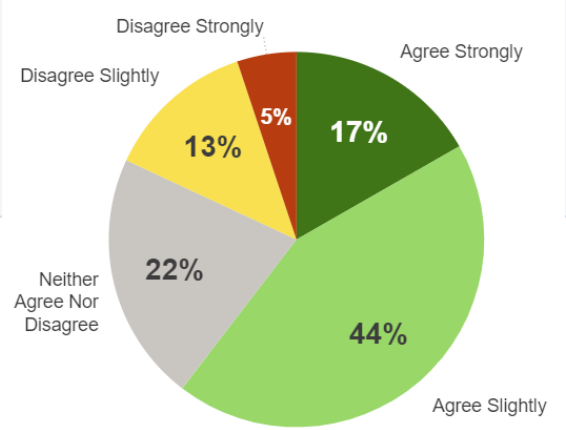
Recommended charts - and when to use each

★ DEFAULT - 100% STACKED BAR

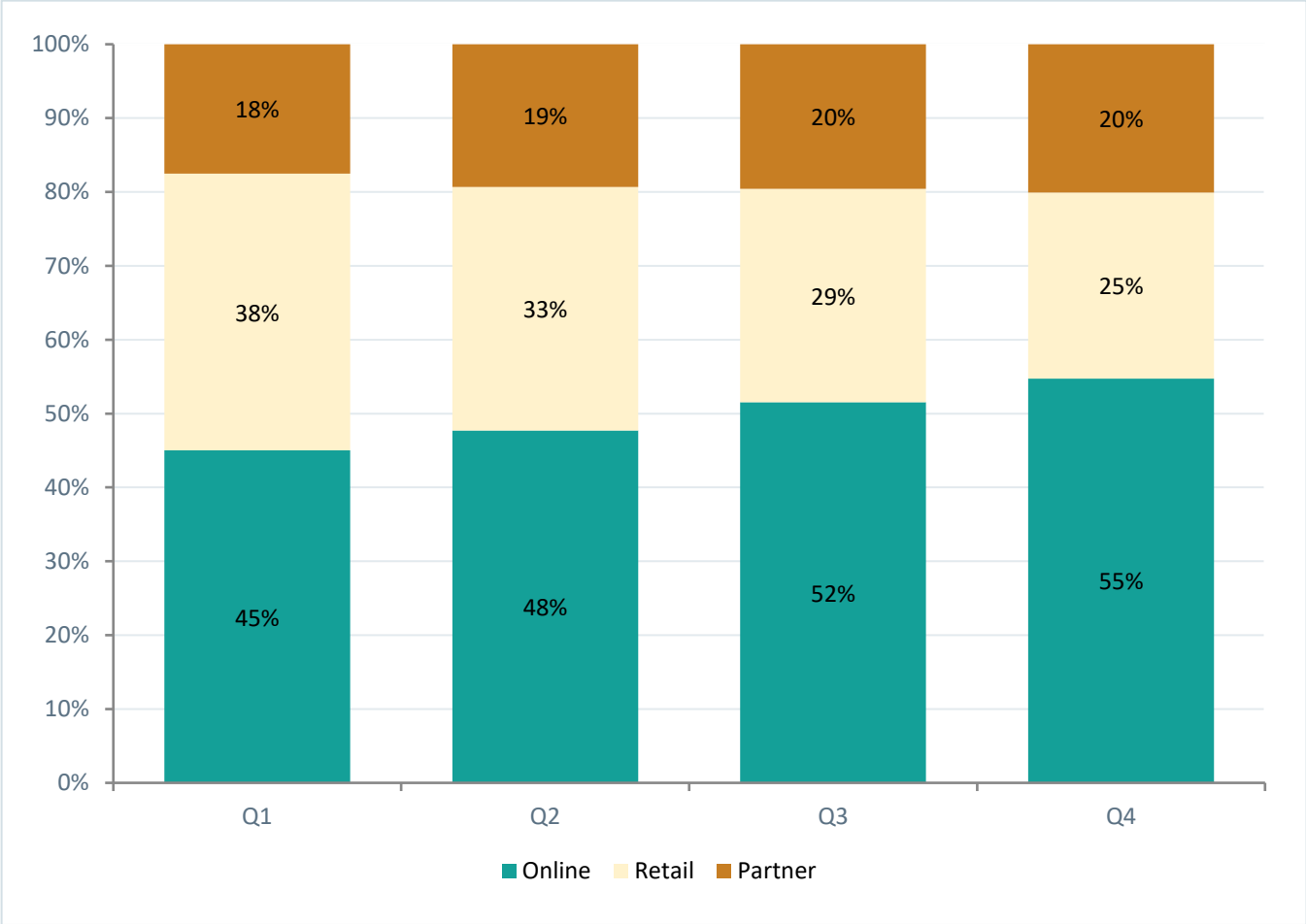
Shows how each category contributes to a whole, with every bar totaling 100%.

ALTERNATIVE - PIE / DONUT

Only with 2–4 parts and large differences. Angle and area are read poorly; label every slice with its share.



Part-to-Whole - example and interpretation



Share of revenue by channel, 100% stacked by quarter

How to read it

Each bar totals 100%. Read one band across bars. Only the bottom band has a stable baseline.

What to look for

Which share grows, which shrinks, and whether the mix is shifting steadily or abruptly.

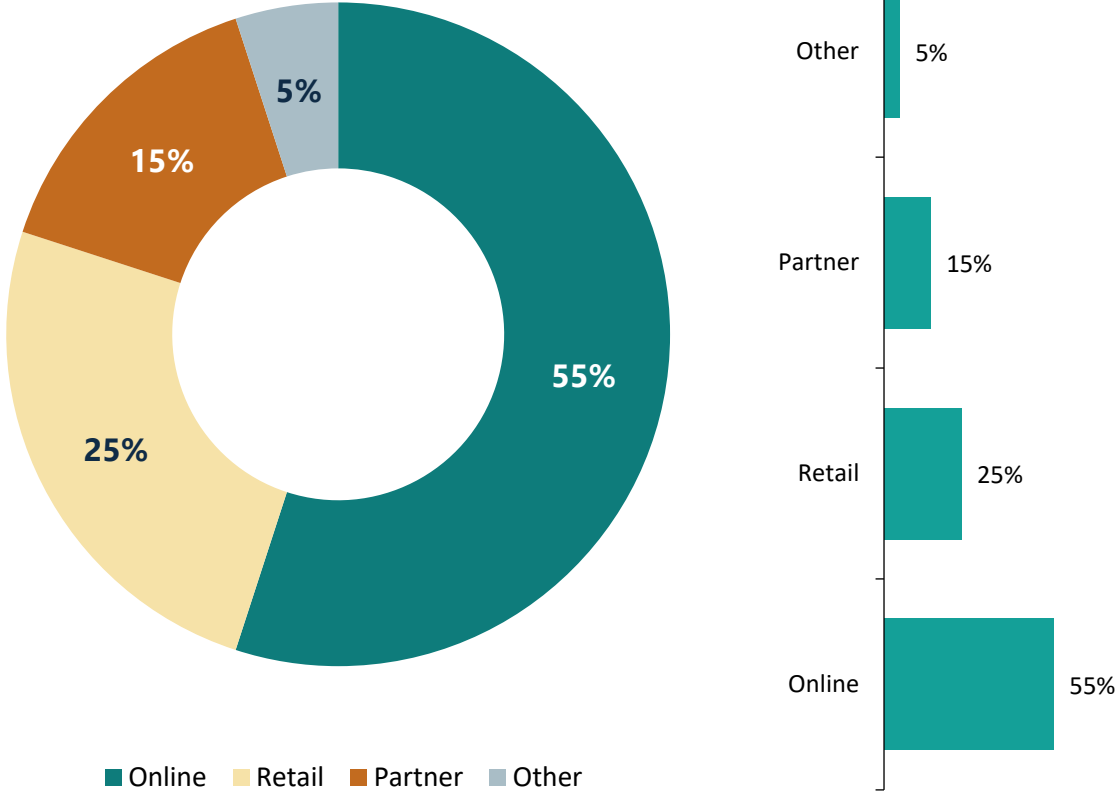
✓ Insight

Online's share of revenue increased from 45% in Q1 to 55% in Q4, while Retail's share declined by 13%.

✕ Not justified

"Retail is shrinking." Retail's *share* fell; in absolute terms it was broadly flat. Share and amount are different claims.

Part-to-Whole – Donut & Bar chart



Same data, two views: donut = the mix; bar = the ranking.

How to read it

Donut: each arc is a slice of one 100% total; read the labels, not the angles.
Bar: read the length of each bar against a common baseline.

Worked example

Q4 revenue = $20M: Online $11M (55%), Retail $5M (25%), Partner $3M (15%), Other $1M (5%).

✓ Insight

Online accounts for 55% of revenue, exceeding Retail and Partner combined (40%), indicating a heavy concentration in a single channel.

✕ Not justified

Partner beats Other by a lot. 15% vs 5% is a 10-point gap.
Use absolute values to contextualize the gap
- Never use a donut for 6+ slices.

🫓 Part-to-Whole - best practices

- Use part-to-whole only when components sum to a meaningful total.
- Label segments directly with percentages.
- Place key comparisons on a stable baseline.
- Show absolute totals when scale matters.
- Use 100% stacked bars to compare composition over time.
- Limit to 3–5 segments; group the rest as “Other.”
- Keep pie charts flat and simple.

💻 Part-to-Whole - implementation in Seaborn and Plotly

Seaborn · static, report-ready

```
import pandas as pd, matplotlib.pyplot as plt,
seaborn as sns

df = pd.DataFrame({
    'quarter': ['Q1','Q2','Q3','Q4'],
    'Online': [180, 210, 250, 300],
    'Retail': [150, 145, 140, 138],
    'Partner': [70, 85, 95, 110]})
   .set_index('quarter')
# Convert revenue to percentage share within each quarter
share = df.div(df.sum(axis=1), axis=0) * 100
sns.set_theme(style='whitegrid')
ax = share.plot(kind='bar', stacked=True,
         figsize=(7,4))
ax.set(xlabel='', ylabel='Share of revenue (%)')
ax.set_title("Online's share rose from 45% to 55%")
plt.show()
```

Plotly · interactive, dashboard-ready

```
import pandas as pd
import plotly.express as px

df = pd.DataFrame({
    'quarter': ['Q1','Q2','Q3','Q4']*3,
    'channel': ['Online']*4 + ['Retail']*4 +
['Partner']*4,
    'revenue': [180,210,250,300, 150,145,140,138,
                70,85,95,110]})

fig = px.bar(df, x='quarter', y='revenue',
 color='channel', barnorm='percent',
 title="Online's share rose from 45% to 55%")
fig.update_layout(yaxis_title='Share of revenue (%)')
fig.show()
```

Seaborn has no stacked-bar API, so we use Pandas' .**plot** on a wide frame.

Plotly takes long (tidy) format and normalises with **barnorm='percent'**.

📐 Purpose 5 - Distribution

Analytical Question

How are the values of one variable spread out?

Recommended charts - and when to use each

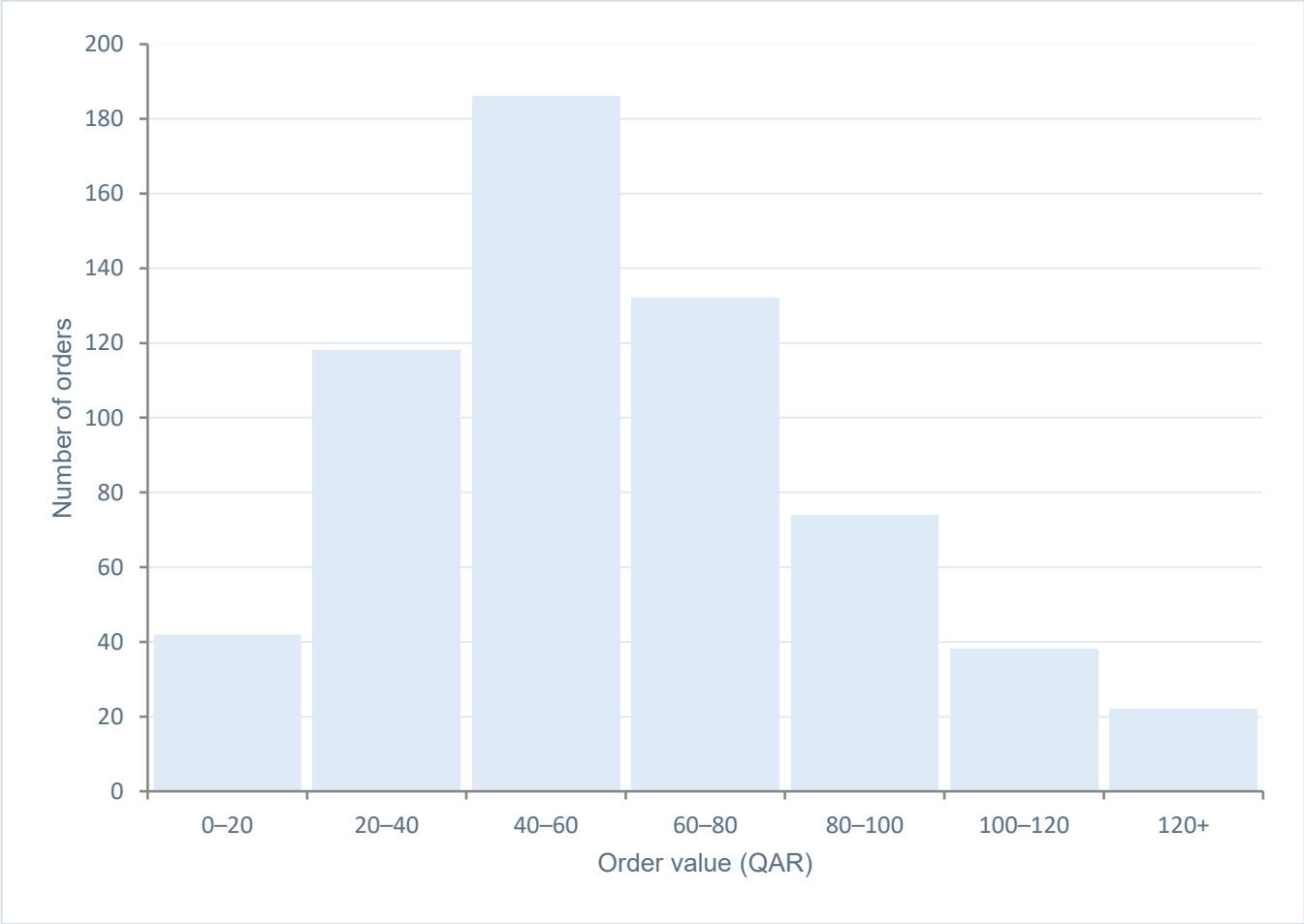
★ DEFAULT - HISTOGRAM

The default for one variable. Shows centre, spread, skew and gaps. The bin width is an analytical choice.

BOX PLOT

Best for comparing distributions across groups compactly. Hides multimodality.

📐 Distribution - example and interpretation



Distribution of order values, 612 orders

How to read it

Height is frequency, not amount. The x-axis is a measurement cut into bins.

What to look for

Centre, spread, skew direction, number of peaks, and a long tail.

✓ Insight

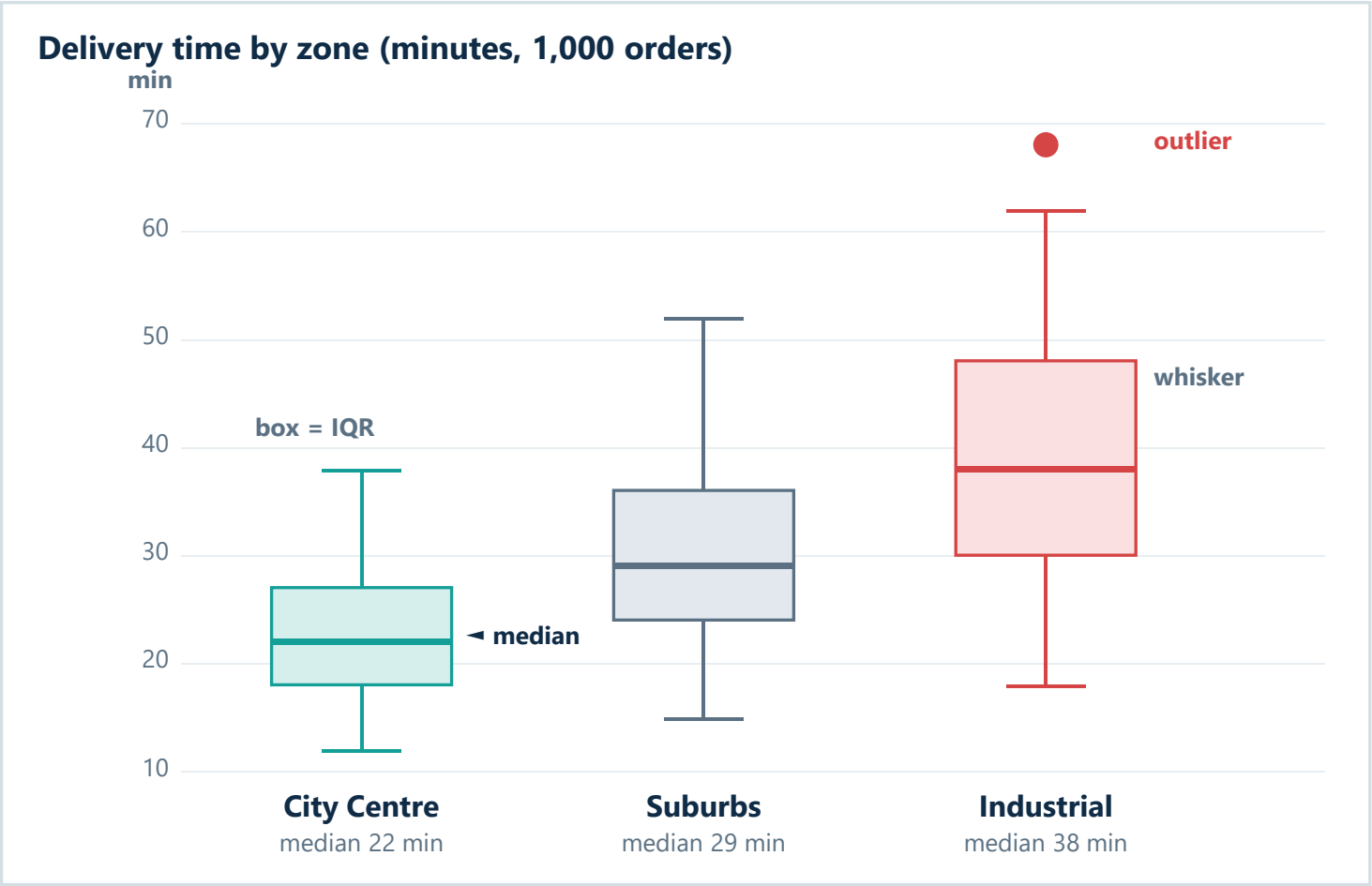
Order values are right-skewed: most orders fall between 20 and 80 QAR, with a thin tail of high-value orders above 120.

✕ Not justified

The average order is 60 QAR, so a typical customer spends 60. With right skew, the mean is pulled up by the tail - report the median.

📐 Box plots - comparing distributions across groups

Question: Is delivery time consistent across zones - and where is it least reliable?



Anatomy of a box

Box = Q1 to Q3 (middle 50%, the IQR). Line inside = median. Whiskers reach the furthest value within 1.5×IQR; dots beyond are outliers.

How to compare groups

Compare medians (typical speed) and box heights (consistency). Heavily overlapping boxes mean the difference is weak.

✓ Insight

City Centre is fastest (median 22 min) and most consistent (IQR 9 min). Industrial is slower (median 38) and far more variable (IQR 18), with a 68-min outlier.

✕ Not justified

"Industrial riders are slower." => not justified

Industrial riders show longer or more variable travel times potentially reflecting route and access constraints.

A box plot shows variation, not cause.

📐 Distribution - best practices

- Test multiple bin widths; too few hide patterns, too many amplify noise.
- Label axes clearly; distinguish frequency from measured values.
- Report the median and IQR for skewed distributions.
- Use histograms for individual distributions and box plots to compare groups.
- Preserve the full range, including outliers.
- State sample size and use consistent bins across comparisons.

💻 Distribution - implementation in Seaborn and Plotly

Seaborn · static, report-ready

```
import pandas as pd, numpy as np, seaborn as sns
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)
orders = pd.DataFrame({
    'value': rng.gamma(shape=3, scale=18,
size=612)})

sns.set_theme(style='whitegrid')
ax = sns.histplot(data=orders, x='value', bins=20,
                  color='#14A098')
ax.axvline(orders['value'].median(),
  color='#D64545', linestyle='--', label='Median')
ax.set(xlabel='Order value (QAR)',
         ylabel='Number of orders')
ax.legend(); plt.show()
```

Plotly · interactive, dashboard-ready

```
import pandas as pd, numpy as np
import plotly.express as px

rng = np.random.default_rng(42)
orders = pd.DataFrame({
    'value': rng.gamma(shape=3, scale=18,
size=612)})

fig = px.histogram(orders, x='value', nbins=20,
         marginal='box',
         title='Order values are right-skewed')
fig.update_layout(xaxis_title='Order value (QAR)',
         yaxis_title='Number of orders')
fig.show()
```

Plotly's marginal='box' stacks a box plot above the histogram, allowing to see both views of the same variable at once.

🔗 Purpose 6 - Relationship

Analytical Question

How are two (or more) variables related?

Recommended charts - and when to use each

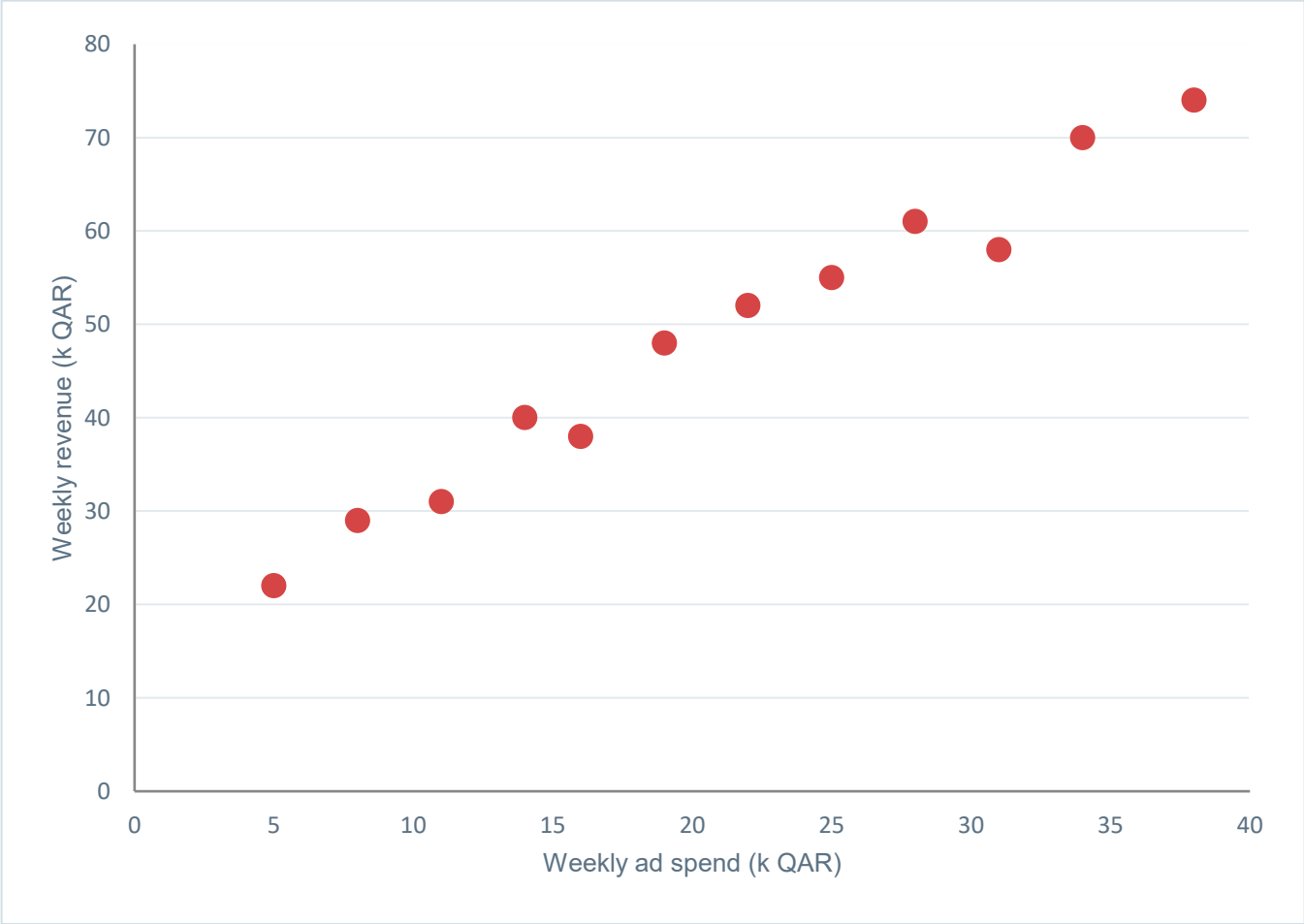
★ DEFAULT - SCATTER PLOT

The default for two numeric variables. Shows direction, strength, form and outliers at once.

ALTERNATIVE - HEATMAP

For a correlation matrix across many variables, as a screening tool before detailed plots.

🔗 Relationship - example and interpretation



Weekly ad spend vs revenue, 12 weeks

How to read it

Each point is one week: x = spend, y = revenue. Read the cloud's shape, not single points.

What to look for

Direction (positive/negative), strength (tight or loose), form (straight or curved), and points far from the pattern.

✓ Insight

"Weeks with higher ad spend tend to show higher revenue; the relationship is positive and fairly strong over this range."

✕ Not justified

"Ad spend causes revenue to rise, so doubling spend doubles revenue." Correlation is not causation, and extrapolating beyond 38k is unsupported.

🔗 Relationship - best practices

- Plot the data before computing a correlation coefficient.
- Put the presumed driver on the x-axis and state the unit of observation.
- Add a trend line only when the form genuinely looks linear.
- Use transparency, jitter or a density plot when points overlap.
- Describe association, not causation, and check for lurking variables.
- Keep conclusions inside the observed data range.

💻 Relationship - implementation in Seaborn and Plotly

Seaborn · static, report-ready

```
import pandas as pd, seaborn as sns
import matplotlib.pyplot as plt

df = pd.DataFrame({
    'ad_spend': [5,8,11,14,16,19,22,25,28,31,34,38],
    'revenue':  [22,29,31,40,38,48,52,55,61,58,70,74]})

sns.set_theme(style='whitegrid')
ax = sns.regplot(data=df, x='ad_spend’,
  y='revenue', color='#D64545', ci=None)
ax.set(xlabel='Weekly ad spend (k QAR)',
       ylabel='Weekly revenue (k QAR)')
ax.set_title('Higher ad spend is associated with
higher revenue')
plt.show()
```

Plotly · interactive, dashboard-ready

```
import pandas as pd
import plotly.express as px

df = pd.DataFrame({
   'ad_spend': [5,8,11,14,16,19,22,25,28,31,34,38],
   'revenue':  [22,29,31,40,38,48,52,55,61,58,70,74]})

fig = px.scatter(df, x='ad_spend', y='revenue',
       trendline='ols',
       title='Ad spend and revenue move together')
fig.update_layout(
    xaxis_title='Weekly ad spend (k QAR)',
    yaxis_title='Weekly revenue (k QAR)')
fig.show()
```

Both add a fitted line in one argument - which is exactly why the wording of the title matters. 'Associated with' is defensible; 'causes' is not.

🗺️ Purpose 7 - Geographic

Analytical Question

Where does the pattern happen, and does location explain it?

Recommended charts - and when to use each

★ DEFAULT - CHOROPLETH MAP

Regions shaded by value.

Use rates per capita, never raw counts - otherwise you are mapping population.

ALTERNATIVE - BUBBLE MAP

Points sized by value.

Better for raw counts and for small areas that a choropleth would hide.

ALTERNATIVE - SORTED BAR BY REGION

Often clearer than a map when you have fewer than ~15 regions and location itself is not the insight.

🗺️ Choropleth Map - example and interpretation



How to read it

Each region is shaded by a value: light-to-dark Blues encode orders per 1,000 residents. Always map a rate, not a raw count.

What to look for

Which markets sit at the dark vs light end, and whether neighbours behave alike. Compare rates, not map area.

✓ Insight

Qatar leads at 7.50 orders per 1k, then Kuwait (6.79) and UAE (6.20). Saudi is moderate (5.00) despite 182k orders; Oman lowest (4.20). Dense city-states buy digitally more often.

✕ Watch out

"Saudi is the weakest market." It has the most orders; its lower rate reflects a dispersed population. Action: Prioritize premium services and express fulfilment in Qatar and the UAE, while expanding last-mile coverage in secondary Saudi cities.

🗺️ Geographic - best practices

- Normalise to a rate: per capita, per household, per 1,000 customers.
- Use a sequential color scale; diverging only around a true midpoint.
- State the geographic unit and the time window.
- Keep projection and bins consistent across comparisons, and disclose the bins.
- Read area averages as areas, not as individuals (ecological fallacy).
- If a bar chart answers the question just as well, use the bar chart.

💻 Geographic - implementation in Seaborn and Plotly

Plotly · interactive, dashboard-ready

```
import pandas as pd
import plotly.express as px

df = pd.DataFrame({
    'iso': ['QAT','ARE','SAU','KWT','OMN'],
    'rate': [7.5, 6.2, 5.0, 6.8, 4.2]})

fig = px.choropleth(df, locations='iso', color='rate',
           color_continuous_scale='Viridis',
           scope='asia',
           title='Order rate per 1,000 residents')
fig.update_layout(coloraxis_colorbar_title='per 1,000')
fig.show()
```

🎯 Purpose 8 - Target / Deviation

Analytical Question

How does actual performance compare with a target or benchmark?

Recommended charts - and when to use each

★ ACTUAL-VS-TARGET BAR

- Bars show the actual value; a marker line shows the target.

- Use when the audience needs both the level reached and whether each unit cleared its bar.

- Watch out: use one consistent target marker, and color bars by met vs missed - not by category.

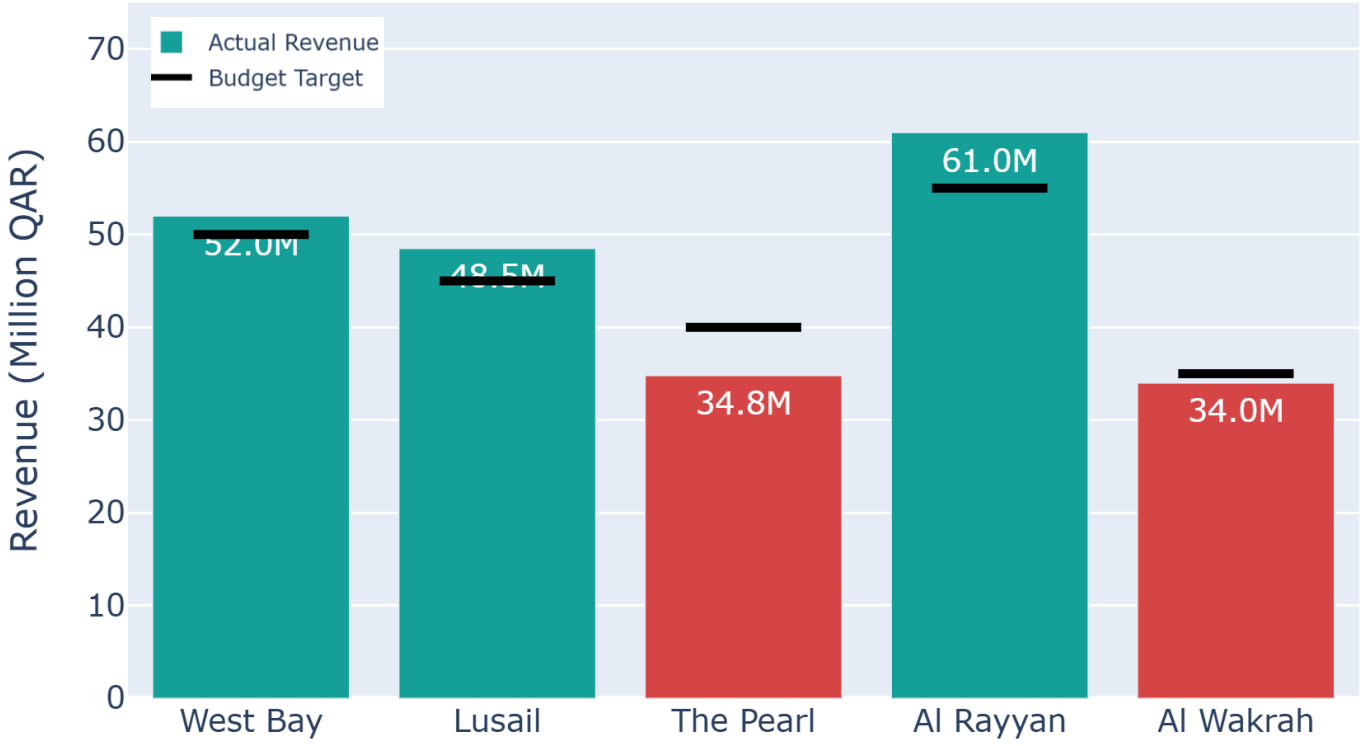
VARIANCE CHART

- Plots the gap itself, above and below a zero baseline.

- Use when the size and direction of the miss is the message, not the absolute level.

- Watch out: always label variance in both absolute units and percent, and sort by variance.

🎯 Actual-vs-Target Bar Chart - example and interpretation



Q4 Revenue Actual-vs-Target Bar Chart

Bars = actual revenue; black tick marks (—) = each zone's budget target. Red bars flag a miss, teal bars a target met.

How to read it

Bars are the **actual** revenue; the black tick is the target. The vertical gap between them is the message.

What to look for

Who is above and below target, how big each gap is, and whether the misses cluster in one area.

Here only The Pearl (34.8M vs 40.0M) falls short.

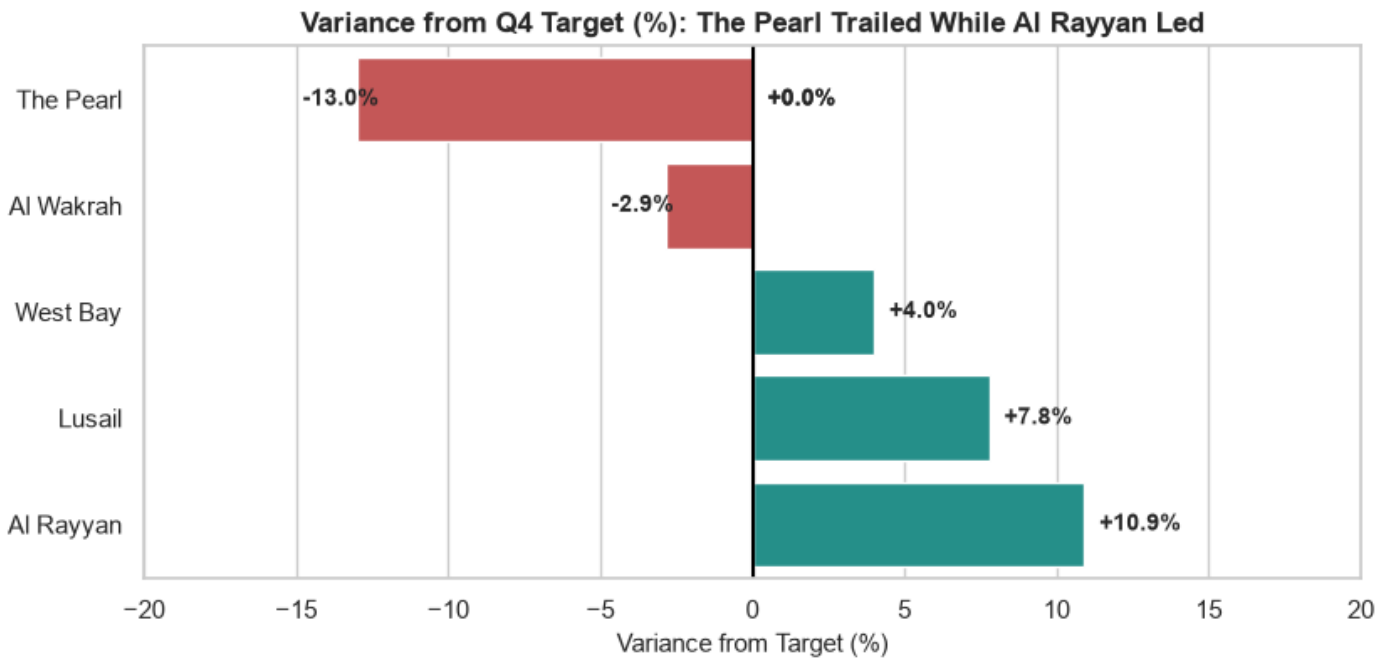
✓ Insight

The Q4 shortfall is localized to The Pearl (-5.2M QAR, -13.0%); every other zone met or beat its target, led by Al Rayyan (+10.9%).

✕ Not justified

"The Pearl underperformed, so its team is failing." The chart shows the gap, not its cause: the target may have been unrealistic or the market shifted.

🎯 Variance Chart - example and interpretation



Q4 Revenue Actual-vs-Target Variance Chart

Variance charts center on zero, removing baseline volume so under- and over-performance can be ranked at a glance.

How to read it

The variance (actual minus target) is plotted left and right of a zero baseline.

Bar direction shows miss or beat; bar length shows how much.

What to look for

Removing the baseline volume puts every zone on one comparable scale, so attention goes straight to the size and direction of each deviation.

✓ Insight

The Pearl (-13.0%) is the only material shortfall, Al Wakrah is marginal (-2.9%), and Al Rayyan (+10.9%) and Lusail (+7.8%) carry the quarter.

✕ Not justified

"The Pearl underperformed, so its team is failing." A variance only quantifies the gap - check target realism, market conditions and scope first.

🎯 Target / Deviation - best practices

- Show the target explicitly alongside the actuals, and state its source (budget, forecast or prior year).
- Use a consistent color rule: teal for met or exceeded, coral red for a shortfall - never color by category.
- Label variance as both an absolute amount and a percentage (e.g., -5.2M QAR, -13.0%).
- Order Variance Chart by variance so the biggest shortfall is impossible to miss.
- Add a tolerance band (e.g., ±2%) so small, normal fluctuations are not read as failures.
- Write the takeaway in the title, and separate the gap from its cause before assigning blame.

💻 Target / Deviation - implementation in Seaborn and Plotly

Seaborn · static, report-ready

```
import pandas as pd, seaborn as sns
import matplotlib.pyplot as plt


df = pd.DataFrame({
    'zone': ['West Bay','Lusail','The Pearl','Al Rayyan','Al
Wakrah'],
    'actual': [52.0, 48.5, 34.8, 61.0, 34.0],
    'target': [50.0, 47.0, 40.0, 55.0, 35.0]})
df['miss'] = df['actual'] < df['target']


sns.set_theme(style='whitegrid')
ax = sns.barplot(data=df, x='zone', y='actual', hue='miss',
                 palette={False:'#14A098', True:'#D64545'},
dodge=False)
ax.scatter(df['zone'], df['target'], marker='_', s=900,
           color='black', label='Budget target')
ax.set(xlabel='', ylabel='Revenue (Million QAR)')
ax.set_title('Only The Pearl missed Q4 target (-13.0%)')
ax.legend(); plt.show()
```

Plotly · interactive, dashboard-ready

```
import pandas as pd
import plotly.graph_objects as go


df = pd.DataFrame({
    'zone': ['West Bay','Lusail','The Pearl','Al Rayyan','Al
Wakrah'],
    'actual': [52.0, 48.5, 34.8, 61.0, 34.0],
    'target': [50.0, 47.0, 40.0, 55.0, 35.0]})
df['var_pct'] = (df.actual / df.target - 1) * 100


fig = go.Figure()
fig.add_bar(x=df.zone, y=df.actual, name='Actual',
            marker_color=['#14A098' if v >= 0 else '#D64545'
                          for v in df.var_pct])
fig.add_scatter(x=df.zone, y=df.target, name='Budget target',
                mode='markers',
                marker=dict(symbol='line-ew', size=28,
                            line=dict(width=3, color='black')))
fig.update_layout(yaxis_title='Revenue (Million QAR)',
    title='Only The Pearl missed Q4 target (-13.0%)')
fig.show()
```

📋 Chart-selection guide: purpose → visualization

Purpose	Analytical question	Recommended
Trend	How does it change over time?	Line chart, area chart
Comparison	How do categories differ?	Bar chart, grouped bar
Ranking	What is highest or lowest?	Sorted horizontal bar
Part-to-Whole	How does each part contribute?	Stacked bar, 100% stacked bar
Distribution	How are values spread?	Histogram, box plot
Relationship	How are variables related?	Scatter plot, heatmap
Geographic	Where does it happen?	Choropleth (rates), symbol map
Target / Deviation	How do we compare with plan?	Bar + target line, bullet chart

⭐ Effective Data Visualization Principles

Choose honestly

- **Fit chart to question -** geometry follows the analytical question; styling cannot rescue the wrong chart type.
- **Title states the finding -** axes carry units, so no verbal explanation is needed.
- **Honest axes and scales -** zero baseline for bars; linear by default.
- **Right grain, right order -** state the aggregation; sort by value unless categories have a natural order.

Reduce noise

- **Color must encode meaning -** highlight, group or scale - otherwise use a single color.
- **Cut clutter, never 3D -** delete gridlines, shadows and duplicate legends.
- **Emphasize one thing -** mute the rest - if everything is emphasized, nothing is.
- **Context and accessibility -** show a benchmark or target; strong contrast, readable type.